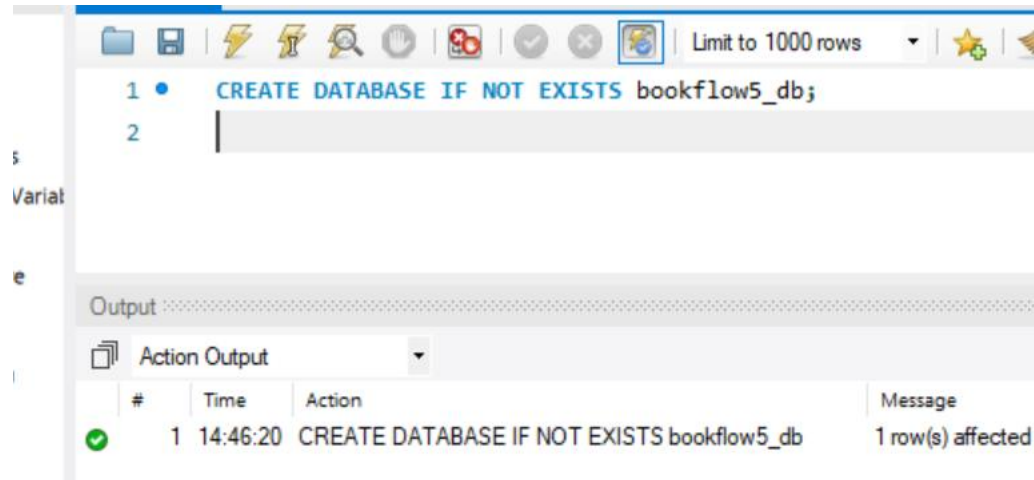


Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

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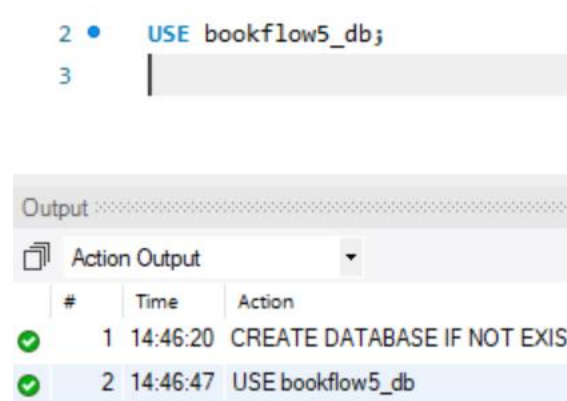
STEP1: CREATE A DATABASE:



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search. The main editor area contains two lines of SQL code: `1 CREATE DATABASE IF NOT EXISTS bookflow5_db;` and `2` (with a cursor). Below the editor, the 'Output' pane is visible, showing a table with columns: #, Time, Action, and Message. The first row indicates the successful execution of the 'CREATE DATABASE IF NOT EXISTS bookflow5_db;' command at 14:46:20, with a message '1 row(s) affected'.

#	Time	Action	Message
1	14:46:20	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected

STEP2: USE DATABASE



The screenshot shows the SQL IDE interface with the 'Output' pane displaying the results of two commands. The first command, 'CREATE DATABASE IF NOT EXISTS bookflow5_db;', was executed at 14:46:20. The second command, 'USE bookflow5_db;', was executed at 14:46:47 and is highlighted in blue. The 'Action' column for the second command is truncated as 'USE bookflow5_db'.

#	Time	Action	Message
1	14:46:20	CREATE DATABASE IF NOT EXISTS bookflow5_db	
2	14:46:47	USE bookflow5_db	

STEP3: CREATE PHYSICIAN TABLE:

```

3 • CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9

```

Output

Action Output

#	Time	Action	Message
✓ 1	14:46:20	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected
✓ 2	14:46:47	USE bookflow5_db	0 row(s) affected
✓ 3	14:48:56	CREATE TABLE physician (employeeid INT PRIMARY ...	0 row(s) affected

STEP4: INSERT DATA INTO PHYSICIAN:

```

10 • INSERT INTO physician VALUES
11 (1,'Viahnavi','Staff Internist','11111111'),
12 (2,'LIKhitha','Attending Physician','22222222'),
13 (3,'divya','Surgical Attending Physician','33333333'),
14 (4,'sushma','Senior Attending Physician','44444444'),
15 (5,'akhila','Head Chief of Medicine','55555555'),
16 (6,'dikshitha','Surgical Attending Physician','66666666'),
17 (7,'archana','Surgical Attending Physician','77777777'),
18 (8,'sri','MD Resident','88888888'),
19 (9,'sai','Attending Psychiatrist','99999999');
20

```

Output

Action Output

#	Time	Action	Message
✓ 1	14:46:20	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected
✓ 2	14:46:47	USE bookflow5_db	0 row(s) affected
✓ 3	14:48:56	CREATE TABLE physician (employeeid INT PRIMARY ...	0 row(s) affected
✓ 4	14:51:56	INSERT INTO physician VALUES (1,'Viahnavi','Staff Inte...	9 row(s) affected

STEP 4 : SELECT

21 • `SELECT * FROM physician;`
 22

	employeeid	name	position	ssn
▶	1	Viahnavi	Staff Internist	111111111
	2	LIKhitha	Attending Physician	222222222
	3	divya	Surgical Attending Physician	333333333
	4	sushma	Senior Attending Physician	444444444
	5	akhila	Head Chief of Medicine	555555555
	6	dikshitha	Surgical Attending Physician	666666666
	7	archana	Surgical Attending Physician	777777777
	8	sri	MD Resident	888888888
	9	sai	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 40 x Apply

Output

Action Output

#	Time	Action	Message
✓ 4	14:51:56	INSERT INTO physician VALUES (1,'Viahnavi','Staff I...	9 row(s) affected Record
✓ 5	14:52:42	SELECT * FROM physician LIMIT 0, 1000	9 row(s) returned

STEP 5: CREATE DEPARTMENT TABLE :

22
 23 • `CREATE TABLE department (`
 24 `departmentid INT PRIMARY KEY,`
 25 `name VARCHAR(100),`
 26 `head INT`
 27 `);`
 28

Output

Action Output

#	Time	Action	Message
✓ 5	14:52:42	SELECT * FROM physician LIMIT 0, 1000	9 row(s) returned
✓ 6	14:54:08	CREATE TABLE department (departmentid INT PRI...	0 row(s) affected

Step : INSERT DATA INTO DEPATMENT :

20

```
29 • INSERT INTO department VALUES
30     (1, 'general', 4),
31     (2, 'psychostry', 7),
32     (3, 'surgery', 9);
33
34 • SELECT * FROM department;
35
```

Result Grid

	departmentid	name	head
▶	1	general	4
	2	psychostry	7
	3	surgery	9
*	NULL	NULL	NULL

department 42 x Apply Revert

Output

Action Output

#	Time	Action	Message
✓ 8	14:57:35	INSERT INTO department VALUES (1, 'general', 4), (...)	3 row(s) affected Records: 3
✓ 9	14:57:39	SELECT * FROM department LIMIT 0, 1000	3 row(s) returned

Step7: INNER JOIN

```

35
36 • SELECT d.name AS Department,
37       p.name AS Head_Physician
38 FROM department d
39 JOIN physician p
40 ON d.head = p.employeeid;
41
42

```

Result Grid

	Department	Head_Physician
▶	general	sushma
	psychostry	archana
	surgery	sai

Result 43 x Read On

Output

Action Output

#	Time	Action	Message
9	14:57:39	SELECT * FROM department LIMIT 0, 1000	3 row(s) returned
10	14:59:15	SELECT d.name AS Department, p.name AS Head_Physician	3 row(s) returned

STEP8: WHERE CLAUSE:

```

42
43 • SELECT *
44 FROM physician
45 WHERE position='Surgical Attending Physician';
46

```

Result Grid

employeeid	name	position	ssn
3	divya	Surgical Attending Physician	333333333
6	dikshitha	Surgical Attending Physician	666666666
7	archana	Surgical Attending Physician	777777777
NULL	NULL	NULL	NULL

Step9: WHERE + LIKE

```

46
47 • SELECT *
48 FROM physician
49 WHERE name LIKE 'likhitha%';
50

```

Result Grid

	employeeid	name	position	ssn
▶	2	LIKhitha	Attending Physician	222222222
*	NULL	NULL	NULL	NULL

physician 46 x Apply Revert

Output

Action Output

#	Time	Action	Message
✓ 12	15:01:07	SELECT * FROM physician WHERE name LIKE 'vais...	0 row(s) returned
✓ 13	15:01:22	SELECT * FROM physician WHERE name LIKE likhit...	1 row(s) returned

STEP 10: COUNT()

```

50
51 • SELECT COUNT(*) AS Total_Physicians
52 FROM physician;
53

```

Result Grid

	Total_Physicians
▶	9

Step11: count distinct :

```

54 • SELECT COUNT(DISTINCT position)
55 FROM physician;
56

```

Result Grid

	COUNT(DISTINCT position)
▶	7

STEP 12: GROUP BY:

```
56
57 • SELECT position,
58       COUNT(*) AS Total_Employees
59 FROM physician
60 GROUP BY position;
61
```

Result Grid	Filter Rows:	Export:	Wrap Cell C
position	Total_Employees		
Staff Internist	1		
Attending Physician	1		
Surgical Attending Physician	3		
Senior Attending Physician	1		
Head Chief of Medicine	1		
MD Resident	1		
Attending Psychiatrist	1		

STEP 13: GROUP BY + HAVING :

```
62 • SELECT position,
63       COUNT(*) AS Total_Employees
64 FROM physician
65 GROUP BY position
66 HAVING COUNT(*) > 1;
67
68
```

Result Grid	Filter Rows:	Export:
position	Total_Employees	
Surgical Attending Physician	3	

STEP14: ORDER BY:

```

67
68 • SELECT *
69 FROM physician
70 ORDER BY name ASC;
71

```

	employeeid	name	position	ssn
▶	5	akhila	Head Chief of Medicine	55555555
	7	archana	Surgical Attending Physician	77777777
	6	dikshitha	Surgical Attending Physician	66666666
	3	divya	Surgical Attending Physician	33333333
	2	LIKhitha	Attending Surgical Attending Physician	
	9	sai	Attending Psychiatrist	99999999
	8	sri	MD Resident	88888888
	4	sushma	Senior Attending Physician	44444444
	1	Viahnavi	Staff Internist	11111111
*	NULL	NULL	NULL	NULL

STEP 15 : ORDER BY DESC:

```

72 • SELECT *
73 FROM physician
74 ORDER BY employeeid DESC;
75
76
77

```

	employeeid	name	position	ssn
▶	9	sai	Attending Psychiatrist	99999999
	8	sri	MD Resident	88888888
	7	archana	Surgical Attending Physician	77777777
	6	dikshitha	Surgical Attending Physician	66666666
	5	akhila	Head Chief of Medicine	55555555
	4	sushma	Senior Attending Physician	44444444
	3	divya	Surgical Attending Physician	33333333
	2	LIKhitha	Attending Physician	22222222
	1	Viahnavi	Staff Internist	11111111
*	NULL	NULL	NULL	NULL

STEP : WHERE + IN

```

78 • SELECT *
79 FROM physician
80 WHERE employeeid IN
81 (
82     SELECT head
83     FROM department
84 );
85

```

Result Grid | Filter Rows: | Edit: |

	employeeid	name	position	ssn
▶	4	sushma	Senior Attending Physician	444444444
	7	archana	Surgical Attending Physician	777777777
	9	sai	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

STEP 17 : SUBQUERY:

```

85 • SELECT *
86 FROM physician
87 WHERE employeeid NOT IN
88 (
89     SELECT head
90     FROM department
91 );
92
93
94

```

Limit to 1000 rows

Result Grid | Filter Rows: | Edit: | Export/Import: |

	employeeid	name	position	ssn
▶	1	Viahnavi	Staff Internist	111111111
	2	LIKhitha	Attending Physician	222222222
	3	divya	Surgical Attending Physician	333333333
	5	akhila	Head Chief of Medicine	555555555
	6	dikshitha	Surgical Attending Physician	666666666
	8	sri	MD Resident	888888888
*	NULL	NULL	NULL	NULL

STEP 18 : JOIN + ORDER BY

```
91 -- );
92 • SELECT d.name,
93        p.name
94 FROM department d
95 JOIN physician p
96 ON d.head=p.employeeid
97 ORDER BY d.name;
98
99
```

Result Grid

	name	name
▶	general	sushma
	psychostry	archana
	surgery	sai

STEP 19 : COUNT + HAVING

```
99 • SELECT position,
100        COUNT(*) AS Total
101 FROM physician
102 GROUP BY position
103 HAVING COUNT(*) >= 2;
104
105
```

Result Grid

	position	Total
▶	Surgical Attending Physician	3

STEP 20: EXISTS

```

104 • SELECT *
105 FROM department d
106 WHERE EXISTS
107 (
108     SELECT 1
109     FROM physician p
110     WHERE p.employeeid=d.head
111 );
112
113

```

Result Grid

	departmentid	name	head
▶	1	general	4
	2	psychostry	7
	3	surgery	9
*	NULL	NULL	NULL

STEP 21: MAX()

```

115
114 • SELECT MAX(employeeid)
115 FROM physician;
116
117
118
119

```

Result Grid

	MAX(employeeid)
▶	9

STEP 22: MIN()

118 • `SELECT MIN(employeeid)`

119 `FROM physician;`

120

121

122

Result Grid



Filter Rows:

	MIN(employeeid)
--	-----------------

1
